

DERMATOLOGY

Bacterial Skin Infections

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LEARNING & TESTABLE OBJECTIVES

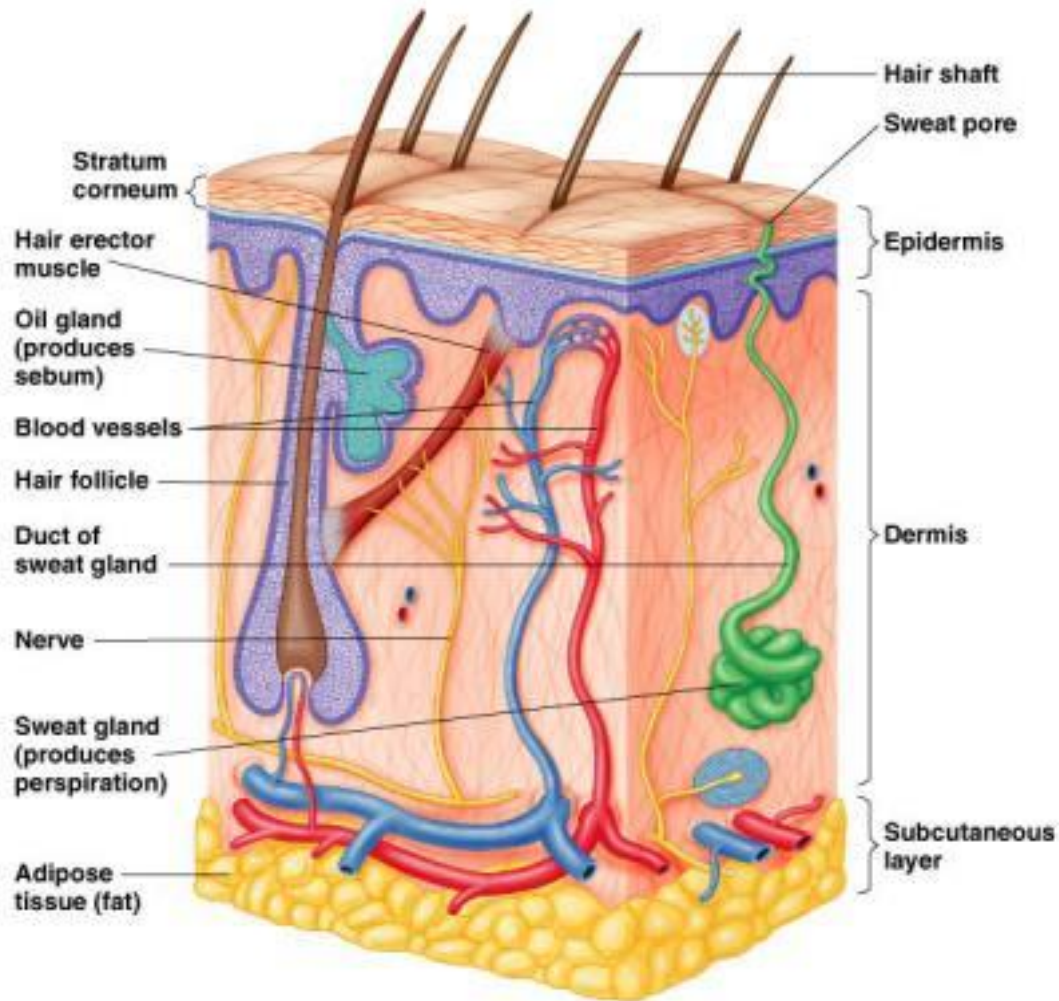
1. List the common normal skin flora and the barriers that protect the skin.
2. Enumerate the various microorganisms involved in different skin infections.

Given case descriptions of skin infections,

3. Identify the etiologic agents and describe their morphological and growth characteristics.
4. Indicate the predisposing/ risk factors for acquiring these infections and modes of transmission.
5. Correlate the virulence factors of the pathogens with the pathogenesis in the host.
6. Identify other sites that may be associated with the pathogen and the possible complications
7. Choose the appropriate clinical sample(s) and laboratory investigations/ tests to confirm the diagnosis.
8. Select the most appropriate antimicrobial agent and explain its mechanism of action.
9. Indicate the general/ specific preventive measures for these infections.

biggest organ and comes into contact with most infections - first line of defense

LAYERS OF THE SKIN AND SOFT TISSUES



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MAJOR MICROBIAL SKIN COLONIZERS

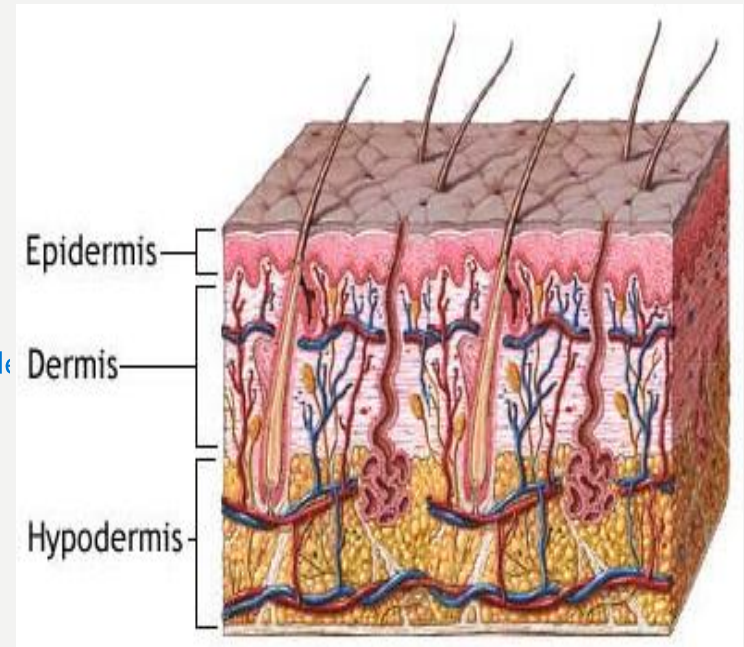
- *Staphylococcus epidermidis*
- *Staphylococcus aureus*
- *Propionibacterium acnes*
- Diphtheroids
(*Corynebacterium spp*)
- Yeasts, Molds
- Rarely, *Pseudomonas*, *E.coli*
etc.

candida

esp in hospitalized pts

HOST BARRIERS TO SKIN INFECTION

- Dense, keratinized outer layer
- Presence of fatty acids
- Acid pH of normal skin
 - pH ~ 5.5
- Exfoliation
- Limited amount of moisture *crevices susceptible*
- Salty sweat *staph aureus/epidermidis*
- Surface temperature ~ 33°C
- Competition
 - Between different species of normal flora
 - Between normal flora and pathogens



Alteration of these factors upsets the ecological balance of the commensal flora and predisposes to infection.

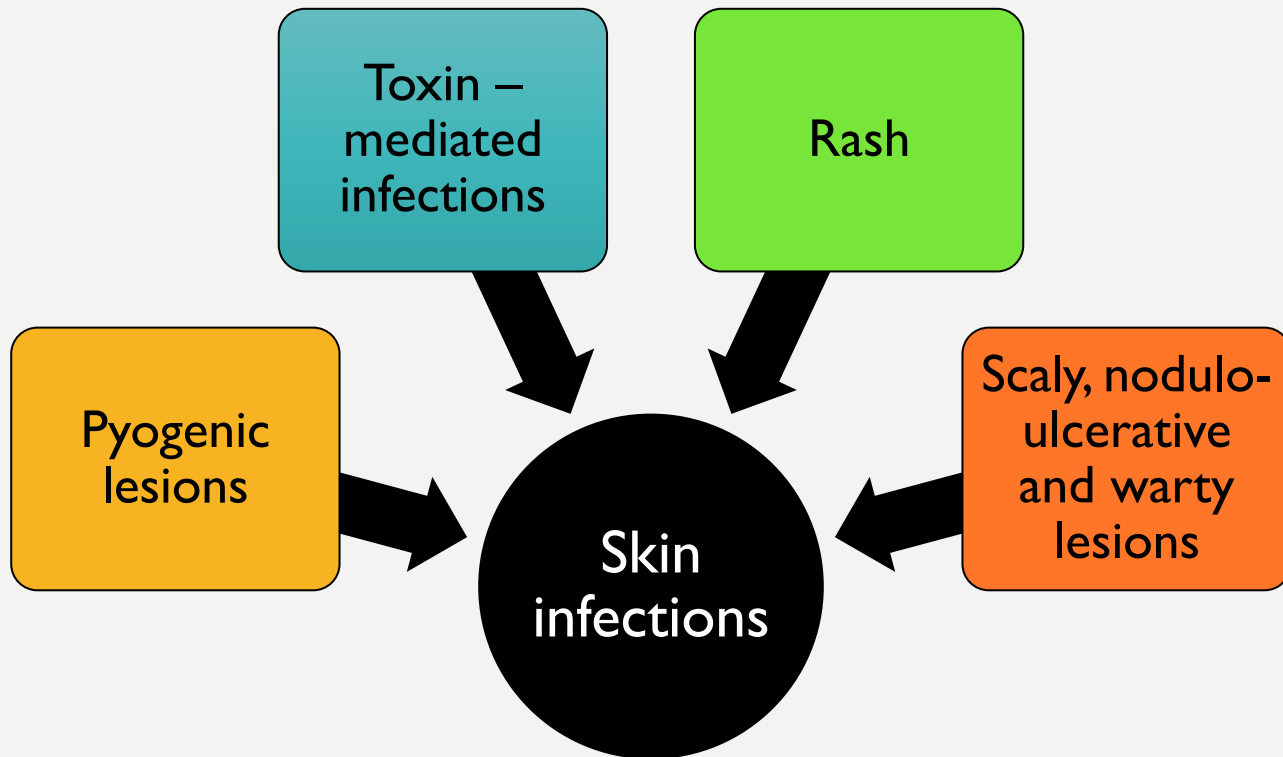
BUG list for Skin Infections

Bacteria	Fungi	Parasites
<i>Staphylococcus aureus</i>	<i>Malassezia furfur</i>	<i>Leishmania tropica</i>
<i>Streptococcus pyogenes</i>	<i>Candida spp</i>	Creeping eruption and Cutaneous Larva Migrans
<i>Propionibacterium acnes</i>	Dermatophytes: <i>Trichophyton</i> <i>Microsporum</i> , <i>Epidermophyton</i>	Filarial worms - <i>Wuchereria</i> and <i>Brugia</i>
<i>Pseudomonas aeruginosa</i>	<i>Sporothrix schenckii</i>	<i>Dracunculus medinensis</i>
<i>Bacillus anthracis</i>	Dematiaceous fungi	<i>Sarcoptes scabiei</i>
<i>Mycobacterium leprae</i>	<i>Blastomyces dermatitidis</i>	<i>Loa loa</i> , <i>Oncocerca volvulus</i>
Atypical mycobacteria (<i>M. marinum</i> , <i>M. ulcerans</i> , <i>M. chelonae</i> , <i>M. fortuitum</i> , <i>M. scrofulaceum</i>)	<i>Coccidioidomyces immitis</i>	<i>Acanthamoeba</i> ↓ Eye infections
<i>Actinomyces</i> and <i>Nocardia</i>		
<i>Borrelia burgdorferi</i>		
<i>Rickettsia rickettsii</i>		
<i>Neisseria meningitidis</i>		
<i>Leptospira interrogans</i>		

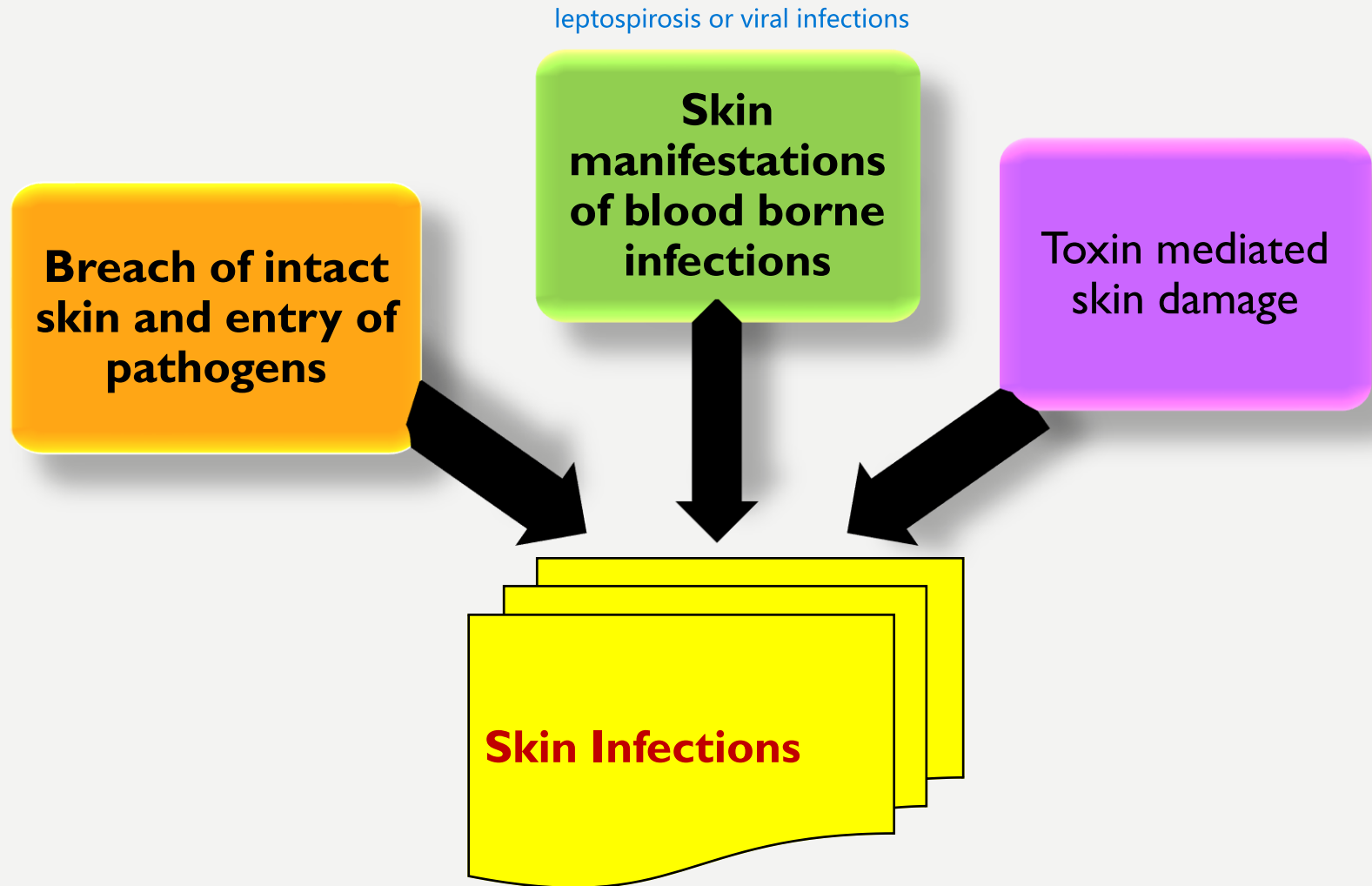
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Different types of skin lesions and underlying mechanisms

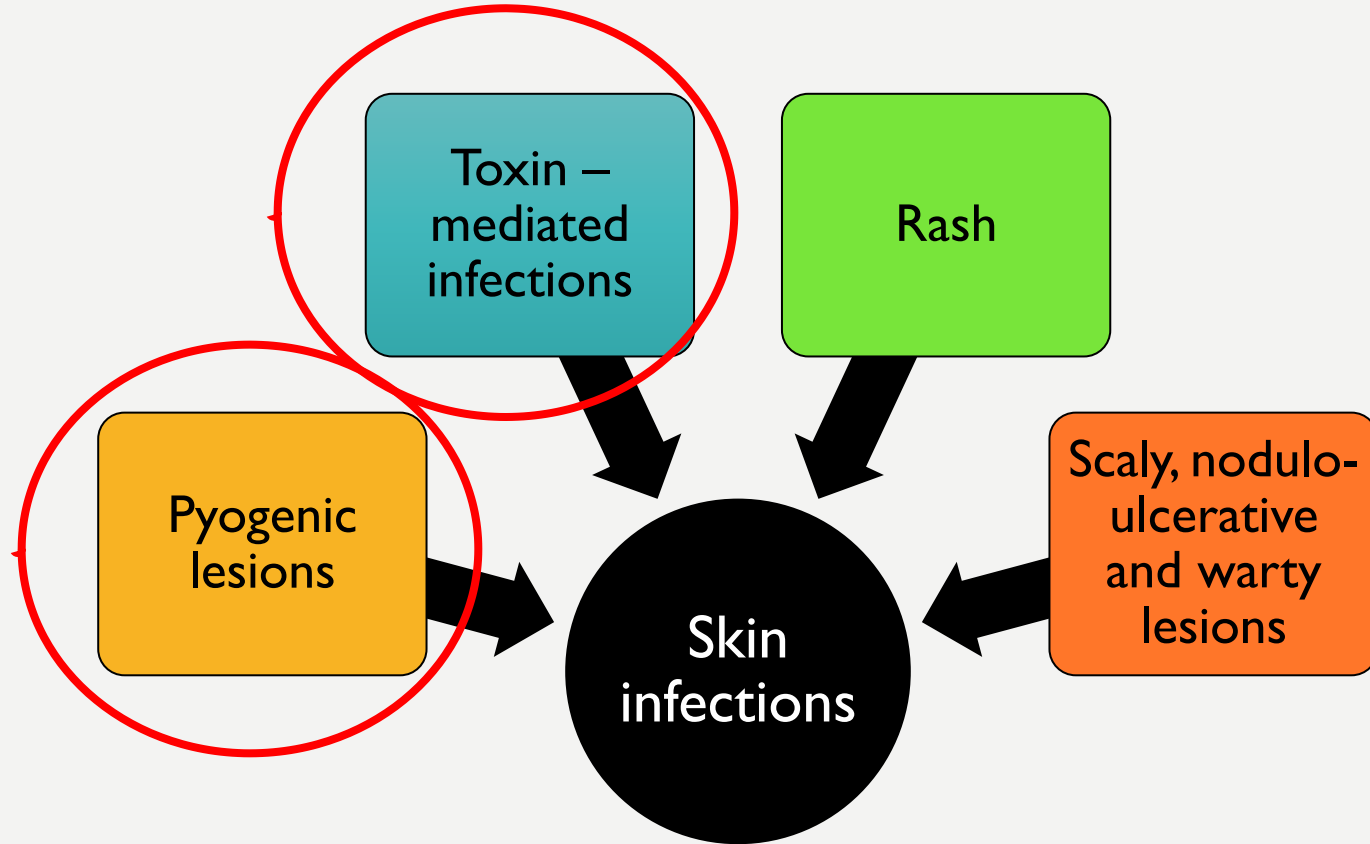
TYPES OF SKIN INFECTIONS



BASIC MECHANISMS OF SKIN INFECTIONS



TYPES OF SKIN INFECTIONS



PYOGENIC SKIN INFECTIONS

Folliculitis

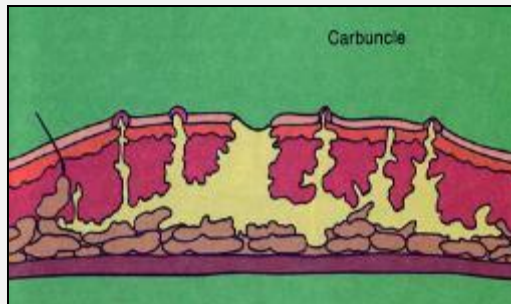
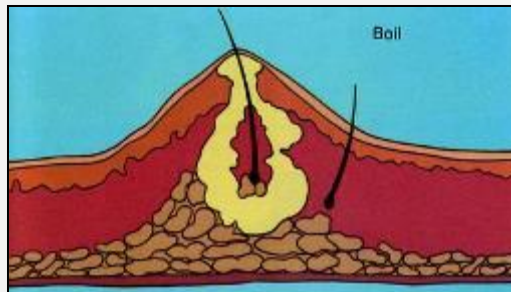


- Due to traumatic implantation of organisms around the hair follicle
- Trauma due to friction and rubbing/ cuts

Furuncle (boil)



Lesion extends beyond the hair follicle to produce a raised and painful lesion



Carbuncle



Several furuncles coalesce together to form a carbuncle. They are seen especially in diabetics.

Important organisms involved

- Gram positive-*Staphylococcus aureus*, *Staphylococcus epidermidis*, Gram negative-*Pseudomonas aeruginosa* (Hot tub folliculitis)

PYOGENIC SKIN INFECTIONS

Abscess

like furuncle but bigger

Painful, fluctuant, red, tender swelling, on which may rest a pustule



- **Most common cause-** *Staphylococcus aureus*.
- *Staphylococcus aureus* is the most common cause of breast abscess in lactating women.
- Other causes of deep seated abscesses are *E.coli*, *Pseudomonas* / mixed infections/ non sporing anaerobes *Prevotella* and *Porphyromonas*

Impetigo

Pustular lesions with dried discharge forming **honey colored crust/ old varnished appearance** on an **erythematous base**.

2 Types: Non bullous (pyogenic) and Bullous (toxin mediated)



Predisposing factors:

- **Seen more in children**
- **Overcrowding**
- **Poor personal hygiene**

Common causes

1. *Staphylococcus aureus*
2. *Streptococcus pyogenes*

can also have mixed infections

PYOGENIC SKIN INFECTIONS

Cellulitis



- ◆ **Acute spreading inflammation of the skin and subcutaneous tissues** with **indistinct edges** with regional lymph node enlargement, malaise, fever and chills.
- ◆ Usually originates from skin lesions or following trauma
- ◆ May lead to bacterial invasion of blood

- Most important cause of cellulitis is *Streptococcus pyogenes*.

Others are,

- *S. aureus* (though usually it causes well circumscribed boils/ abscesses)
- *Pseudomonas*- cellulitis of foot and also osteomyelitis especially in diabetic patients due to colonization of shoe soles.
- *Vibrio vulnificus* can cause cellulitis and blisters in fishermen who get exposed to sea water
- Violaceous lesions caused by *Erysipelothrix rhusiopathiae*.

Note: Some of these will be discussed in 'Wound Infections' lecture

PYOGENIC SKIN INFECTIONS

Erysipelas



An acute deeper infection in the dermis and superficial lymphatics.

Pyogenic and toxin mediated infection.

Salmon red rash, inflamed area with well demarcated edges, lymphadenopathy

Causative agent-
Streptococcus pyogenes

Erysipeloid/ Whale finger/ Seal finger



Violaceous lesions developing in wounds caused due to fishing / handling whales/ seals/ pigs.(Occupational disease)

Causative agent-
Erysipelothrix rhusiopathiae

Details will be discussed in
'wound infections' lecture

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TOXIN-MEDIATED SKIN INFECTIONS

Staphylococcal Scalded Skin Syndrome and Bullous impetigo



Diffuse epidermal peeling due to exfoliatin



extensive peeling

Caused by strains of *Staphylococcus aureus* that form the **toxin, Exfoliatin/Exfoliative exotoxin**.

Mode of action of the toxin

They are serine proteases that split the intercellular bridges in the Stratum granulosum of epidermis.

Act specifically on the **cadherin desmoglein 1**, an adhesion protein in the desmosomes of the Stratum granulosum



Toxic Shock Syndrome

- Severe **headache, sore throat, fever** as high as 105° F (40.6° C), **rashes** that spread from the face to the rest of the body, **palms and soles** and undergo desquamation; dehydration and watery **diarrhea** leading to **shock** and death.



Predisposing factors:

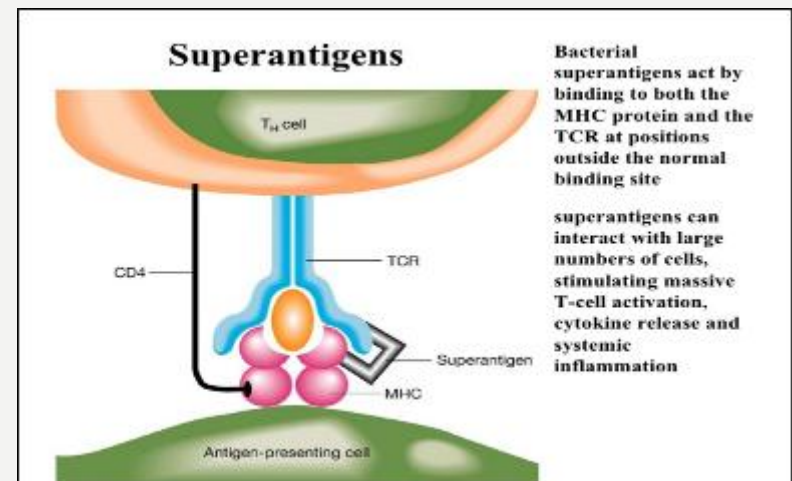
Infected tampons.

Infected dressings/ surgical packing/ sutures.

Causative agents and toxins

- Staphylococcus aureus*- TSST-1 toxin
- GpA Streptococcus pyogenes*-
Streptococcal pyrogenic exotoxin

all from cytokine storm



Toxin: TSST-1 is a superantigen

Scarlet fever



Clinical features- **Strawberry tongue, circumoral pallor, Sandpaper rash—blanching, sparing of palms and soles**
Caused by **Streptococcus pyogenes (Gp A Strep)**

no widespread cytokine reaction

Streptococcal Toxic Shock Syndrome

- **Similar to staphylococcal TSS.**
- Predisposing factors: Immuno-compromised patients such as HIV pts., diabetics, patients on anticancer therapy
- Strains produce **SPE toxin**
- May be associated with Necrotizing Fasciitis

palms and soles involved in TS

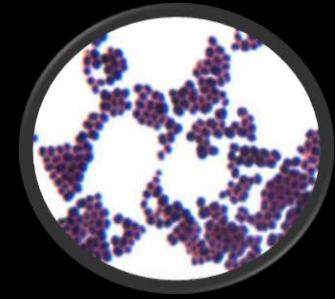
Necrotizing fasciitis

Most common agent- *Strep pyogenes*

(Details will be discussed in 'wound infections' lecture)



BACTERIA CAUSING SKIN INFECTIONS



Staphylococcus aureus

Diseases caused by *S.aureus* (The Big Picture)

Systemic infections

- Bacteremia and Endocarditis
- Pneumonia and Empyema
- **Osteomyelitis and septic arthritis**



Localized suppurative skin and soft tissue infections

Impetigo, Folliculitis, Furuncle, Carbuncle, Abscess, mastitis, wound infections



Toxin mediated diseases

- Food poisoning
- **Staphylococcal Scalded Skin Syndrome**
- **Toxic Shock Syndrome**

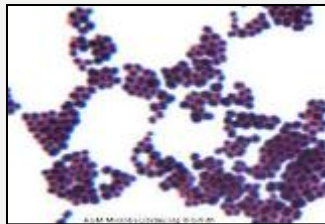
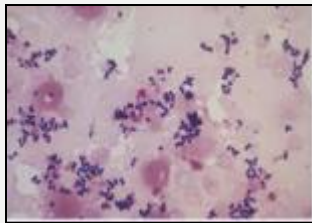


S.aureus

Morphology and Growth characteristics

- Gram positive cocci in clusters.
- Non-motile and non-sporing.

MICROSCOPY



Gram positive cocci
in clusters seen with
inflammatory cells in
a stained smear of
pus

CULTURE



✓ **Nutrient agar:**
Golden yellow opaque
colonies



✓ **Blood agar** shows beta
hemolytic colonies



✓ **Mannitol salt agar** is a selective
medium used for food poisoning
cases. *S.aureus* forms yellow colonies

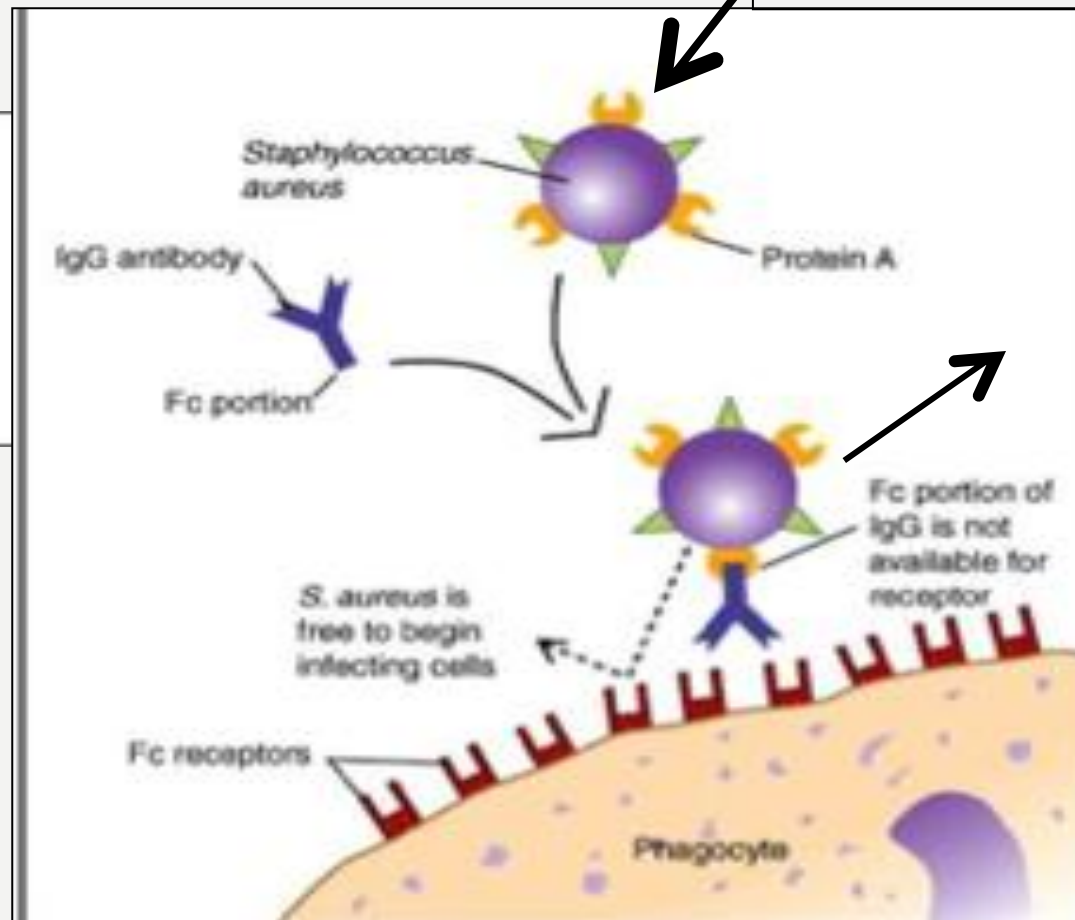
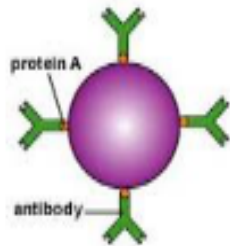
Identification tests

- Catalase positive
 - Coagulase positive
- (*S.epidermidis* is
coagulase negative)

S. aureus Important virulence factors

SURFACE COMPONENT-PROTEIN A

Protein A helps in immune evasion by attaching to Fc portion of Ig, thus evading opsonization

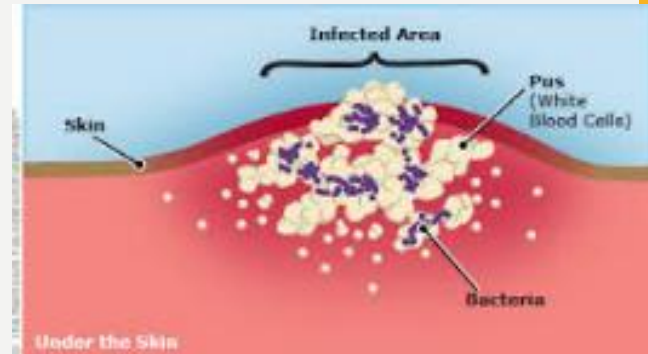


ENZYMES

Biological effects

<u>Coagulase</u>	<u>Converts fibrinogen to fibrin</u>
Fibrinolysin	Dissolves fibrin clots
Lipases	Hydrolyzes lipids (important in chronic furunculosis)
Hyaluronidase	Hydrolyzes hyaluronic acids in connective tissue, promoting the spread of staphylococci in tissue (amount is insignificant compared to hyaluronidase formed by streptococci)

Well circumscribed,
walled off lesions



Coagulase enzyme plays an important role in producing a fibrin barrier and walling off lesions, protecting it from phagocytosis.

Staphylococcal localizing lesions are usually well circumscribed (**abscesses**) as against streptococcal lesions that are spreading in nature (**cellulitis**)



TOXINS	Biological effects
Cytotoxins: Hemolysin; Panton-Valentine Leukocidins	Pore forming toxins which affect cells including erythrocytes and leukocytes.
Exfoliative toxins/ Exfoliatin	Hydrolysis of desmosomal proteins, split the intercellular bridges in the Stratum granulosum of epidermis. Responsible for Staphylococcal Scalded Skin Syndrome
Enterotoxins(A-E)	Responsible for food poisoning
Toxic Shock Syndrome Toxin-1 (TSST-1)	Super-antigen (stimulates widespread activation of T cells and release of cytokines); produces leakage or cellular destruction of endothelial cells

TREATMENT

For uncomplicated infections caused by **Methicillin Sensitive Staphylococcus aureus (MSSA)**

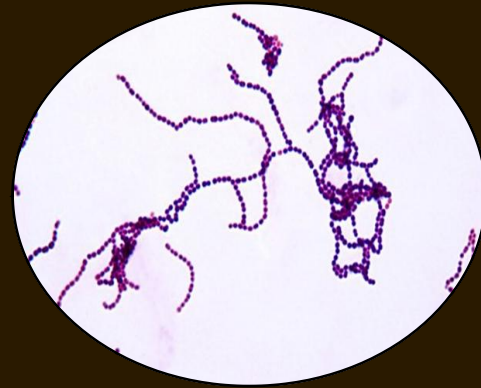
- Cephalexin –cephalosporin/ Amoxycillin+ Clavulanic acid / Ampicillin+ Cloxacillin(or Dicloxacillin)

For **Methicillin Resistant Staphylococcus Aureus (MRSA infections)** antibiotics like Vancomycin, Clindamycin, Linezolid or Rifampin should be used.

MeckA gene

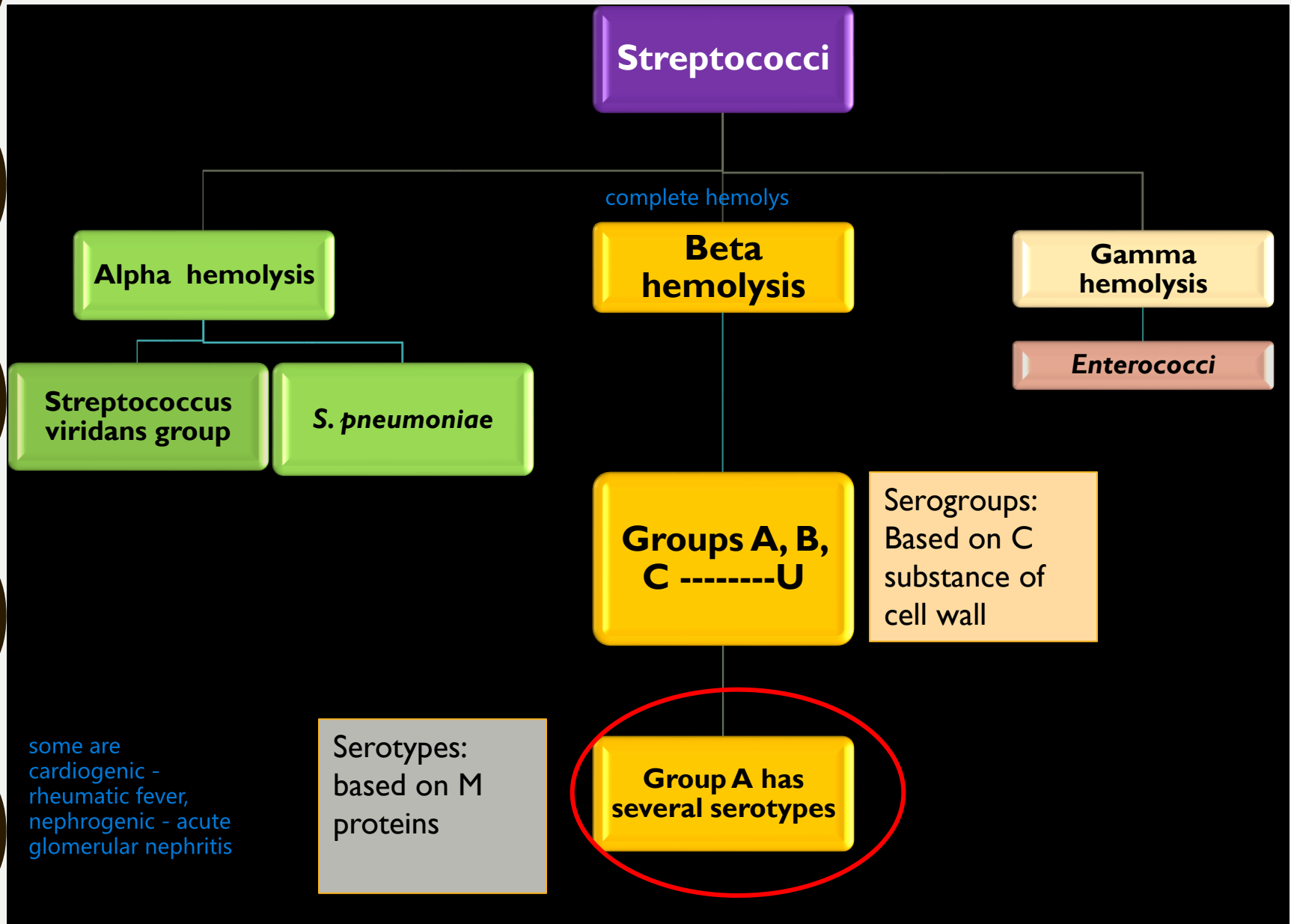
PREVENTION

- **Avoid contaminating areas when there is an active skin lesion caused by *S. aureus***
- **Regular hand washing by hospital personnel helps to break the chain of transmission.**
- Regular screening of health care personnel/ community for MRSA carriage by testing nasal and cutaneous swabs.
- Use of Mupirocin ointment for carrier state. Vancomycin therapy for persistent carriers.



*Streptococcus
pyogenes*

CLASSIFICATION OF STREPTOCOCCI

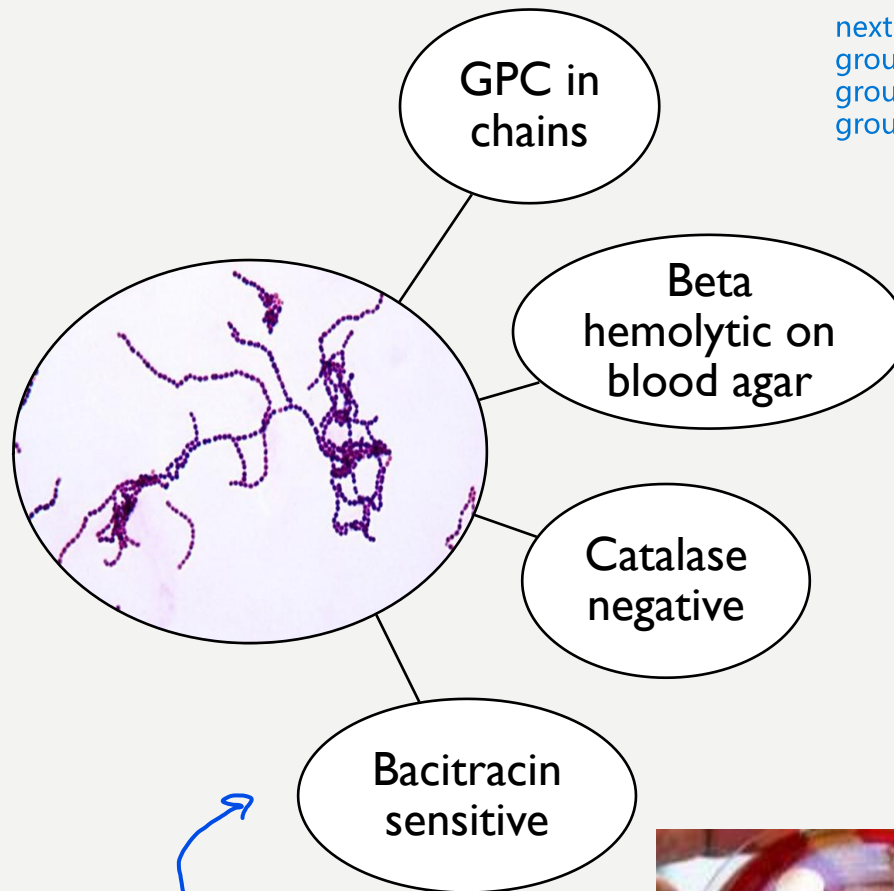


GP A STREPTOCOCCI (GAS)/ STREPTOCOCCUS PYOGENES

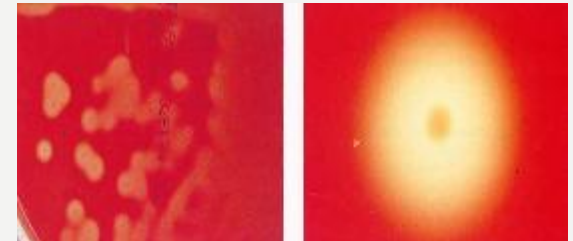
- Pyoderma/ impetigo
- Cellulitis
- Erysipelas
- Scarlet fever
- Necrotizing Fasciitis
- Toxic Shock Syndrome

- Infection acquired from other people resulting in colonization of the skin.
- Bacteria enter through minor breaks in the epithelium
- Increased risk in patients with lower socioeconomic status, very young and the elderly and diabetics.

STREPTOCOCCUS PYOGENES (GPA STREPTOCOCCI): MORPHOLOGY AND IDENTIFICATION



next in alphabet for identifying test
group A - bacitracin
group B - cam test
group D - escalin



how you know it's strep group A



VIRULENCE FACTORS OF S.PYOGENES

★ Streptococcal Pyrogenic Exotoxins (SPE) A-C Pyrogenic/Erythrogenic toxin

Produced by lysogenized strains and acts as a Superantigen: TSST-1

Responsible for the diffuse **erythematous rash of scarlet fever** (that may be seen with pharyngitis)

★ **Hyaluronidase** : hydrolyses ground substance

Streptokinase: breaks fibrin clots

Streptococcal DNase : liquefies pus

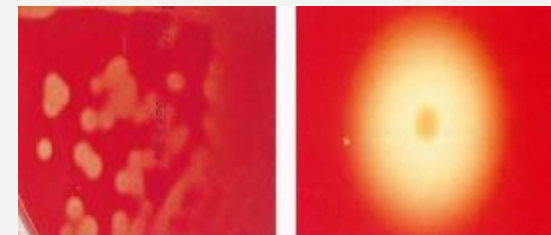
Responsible for cutaneous lesions that are **spreading and invasive in nature-Cellulitis**

C5a peptidase

Evasion of Phagocytosis

Streptolysin O : Oxygen labile, pore forming cytotoxin , antigenic

Streptolysin S is Oxygen stable ,nonimmunogenic



Non suppurative complication of streptococcal skin infections: **Acute Glomerulonephritis**

- A disease characterized by the sudden appearance of edema, hematuria, proteinuria, and hypertension.
- Acute post-streptococcal glomerulonephritis (APSGN) results from an antecedent infection of the **skin (impetigo)** or throat (pharyngitis) caused by nephritogenic strains of group A beta-hemolytic streptococci.
- **Specific M types, such as 49, 55, 57, and 60, are most commonly associated with skin infections and can cause AGN (Nephritogenic strains)**
- Common in children between 6-10 yrs of age. May be seen in adults of any age.
- Appears 1-4 weeks after a streptococcal infection of the skin. Due to the deposition of Immune complexes on Glomerular Basement Membrane (GBM) and complement activation  **Type III hypersensitivity**).
- **Diagnosed by Anti-DNAse B detection**

rather than anti-streptolysin-O

LAB DIAGNOSIS, TREATMENT

LAB DIAGNOSIS

- Samples: Exudate/ pus/ swab
- Microscopy-GPC in chains
- Culture- Beta hemolysis
- Tests for identification-Catalase-neg, Bacitracin-sensitive
- Antibiotic Sensitivity Test

TREATMENT

- Penicillin
- Or
- Amoxycillin
- Or
- Cephalexin

Propionibacterium acnes

- Obligate anaerobic gram-positive rod
- Normal flora of the skin
- Most common anaerobe isolated in the lab
- Causes acne vulgaris



*Bacillus
anthracis*

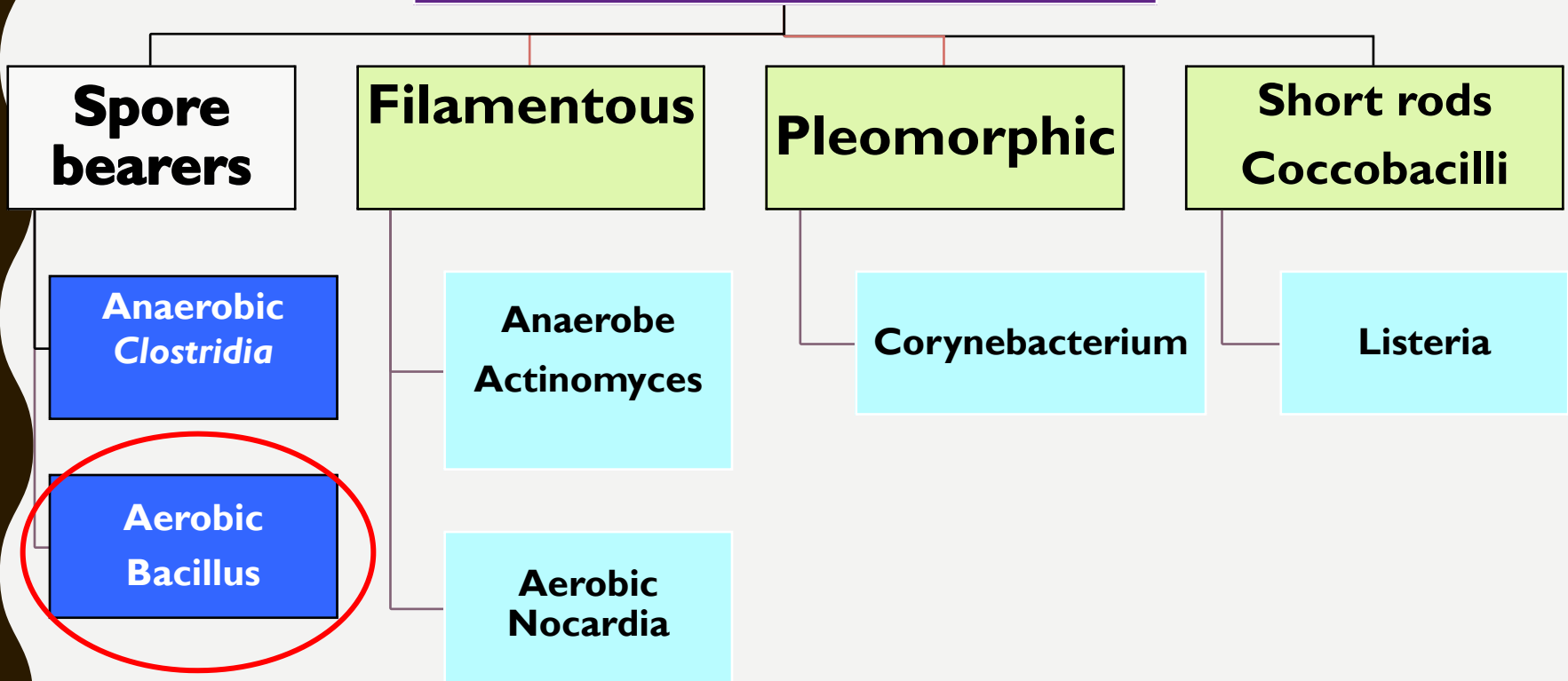
CASE HISTORY

- A 32-yr-old butcher, from Zambia (Africa), visits the local physician with a skin lesion and severe swelling of his right arm. He notes that a few days ago the lesion was a small painful pimple, which rapidly progressed into a large ulcer.
- O/E, the lesion is ulcerative, 10 mm in diameter and has a black necrotic appearance (see image). The surrounding tissue is severely edematous and there is painful lymphadenopathy.

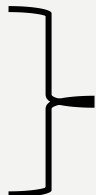


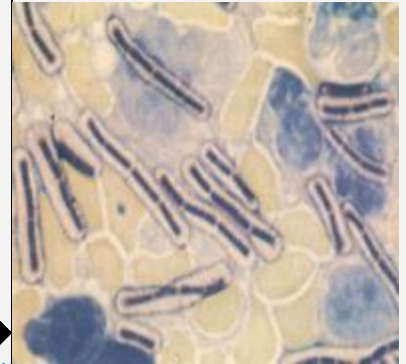
- What is the likely condition? What is the etiologic agent?
- Recollect an instance in the US history when this microorganism was used as a weapon for bioterrorism?
- What virulence factors are most likely responsible for the cutaneous lesions?

Gram Positive Rods



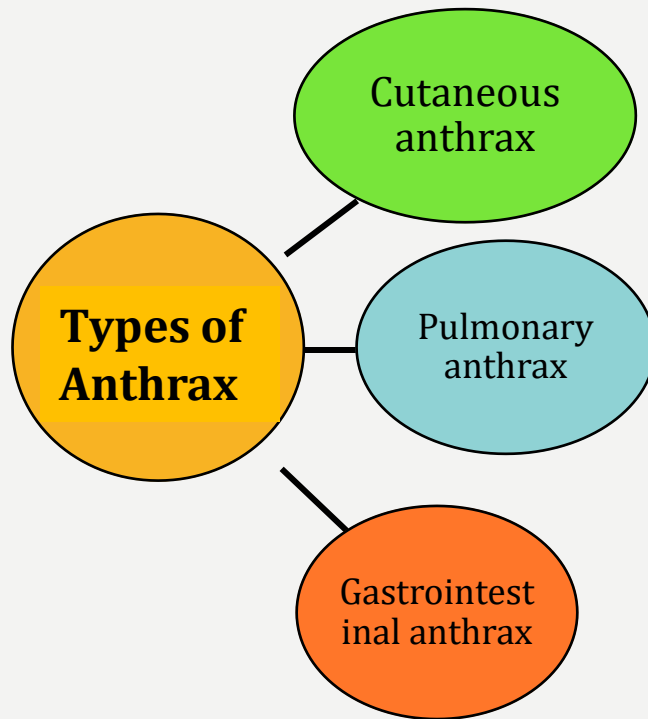
BACILLUS ANTHRACIS

- Ubiquitous / zoonotic, lives in soil
- Causes anthrax in animals and humans
- Large **Gram positive bacilli**
- Arranged in **long serpentine chains with a 'jointed bamboo stick' appearance or 'Box car shaped' morphology.**
- Forms **endospores** (not seen in clinical samples) only in culture media
- Non-motile
- Has a polypeptide capsule made up of poly-d glutamic acid.
- Culture- Blood agar: non hemolytic, large, irregular colonies with raised edge, '**MEDUSA HEAD**' appearance.
- **Exotoxins- Anthrax toxin**
- **Tripartite toxin** 
 - Protective antigen**
 - Edema factor**
 - Lethal factor**



EPIDEMIOLOGY

- Cutaneous anthrax is common in leather industry workers and butchers.
- Pulmonary anthrax- Bioterrorism- Classified as a 'Category A' biological agent



Exposure to spores present in contaminated soil or in infected animal products (hide, hair and wool)

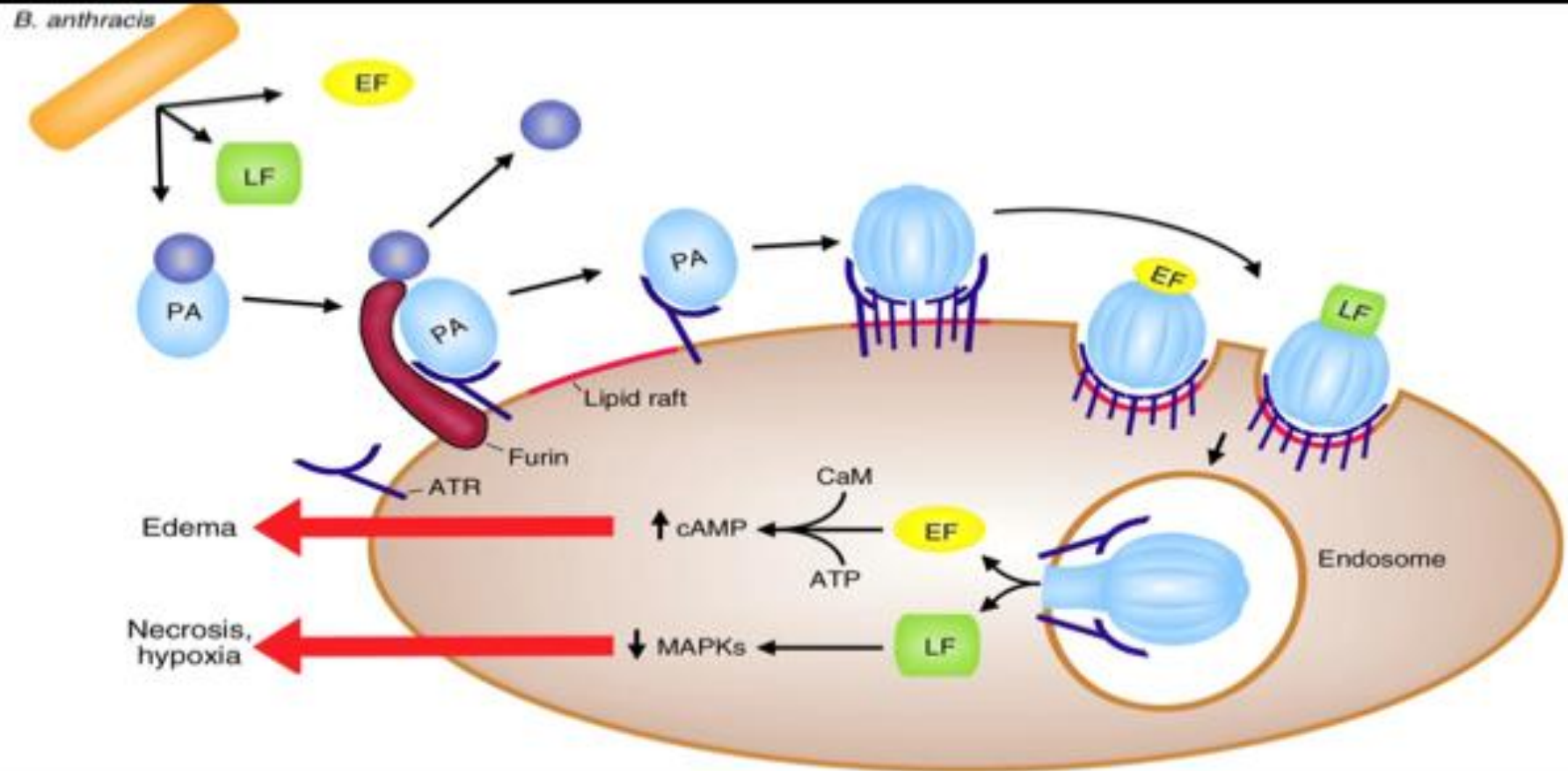
AKA **Hide Porter Disease/ Malignant pustule**

Inhalation of dust , filaments of wool
AKA **Wool Sorter's Disease**

Consumption of infected meat

All types may lead to **septicaemia and meningitis**

Mode of Action of Anthrax Toxin



- **PA = protective antigen:** required for cell entry of EF and LF
- **EF = edema factor:** adenylate cyclase -intracellular cAMP -efflux of fluids and ions leading to **edema**
- **LF = lethal factor:** mitogen-activated protein kinase kinase (MAPKK) **metalloprotease** -disrupts cell signaling -**cell death & tissue necrosis**

CUTANEOUS ANTHRAX

- **ANTHRAX**: Name derived from anthracite= COAL(due to **black lesions of skin called eschar**).
- Entry of spore through abraded skin. Face, neck, hands, back are affected.



painless, pruritic papule → vesicle → necrotic ulcer covered by a characteristic **black eschar**

Diagnosis and Treatment

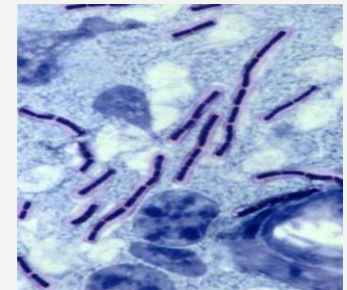
From skin lesions

Gram stain: **GPR** ("**box car chains**")

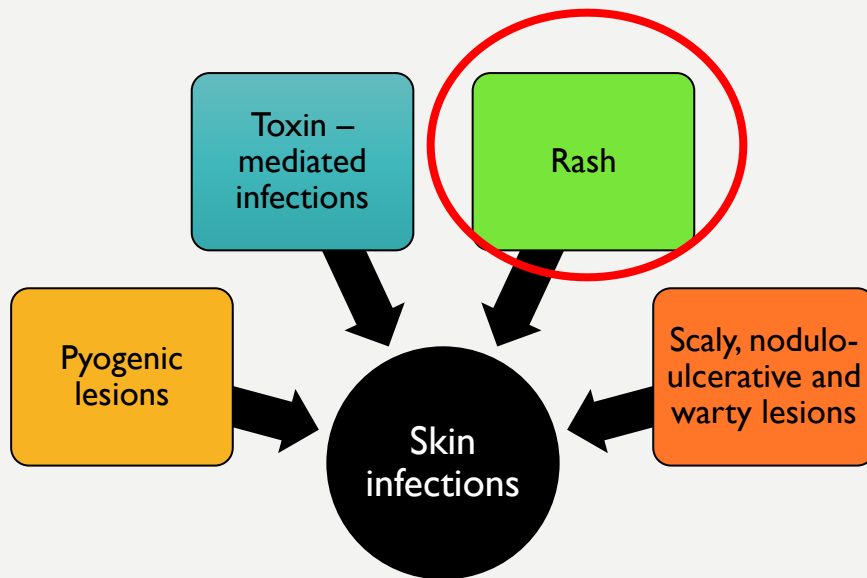
Culture: Large adherent colonies; nonhemolytic colonies

Treatment

Ciprofloxacin (fluoroquinolones)



TYPES OF SKIN INFECTIONS



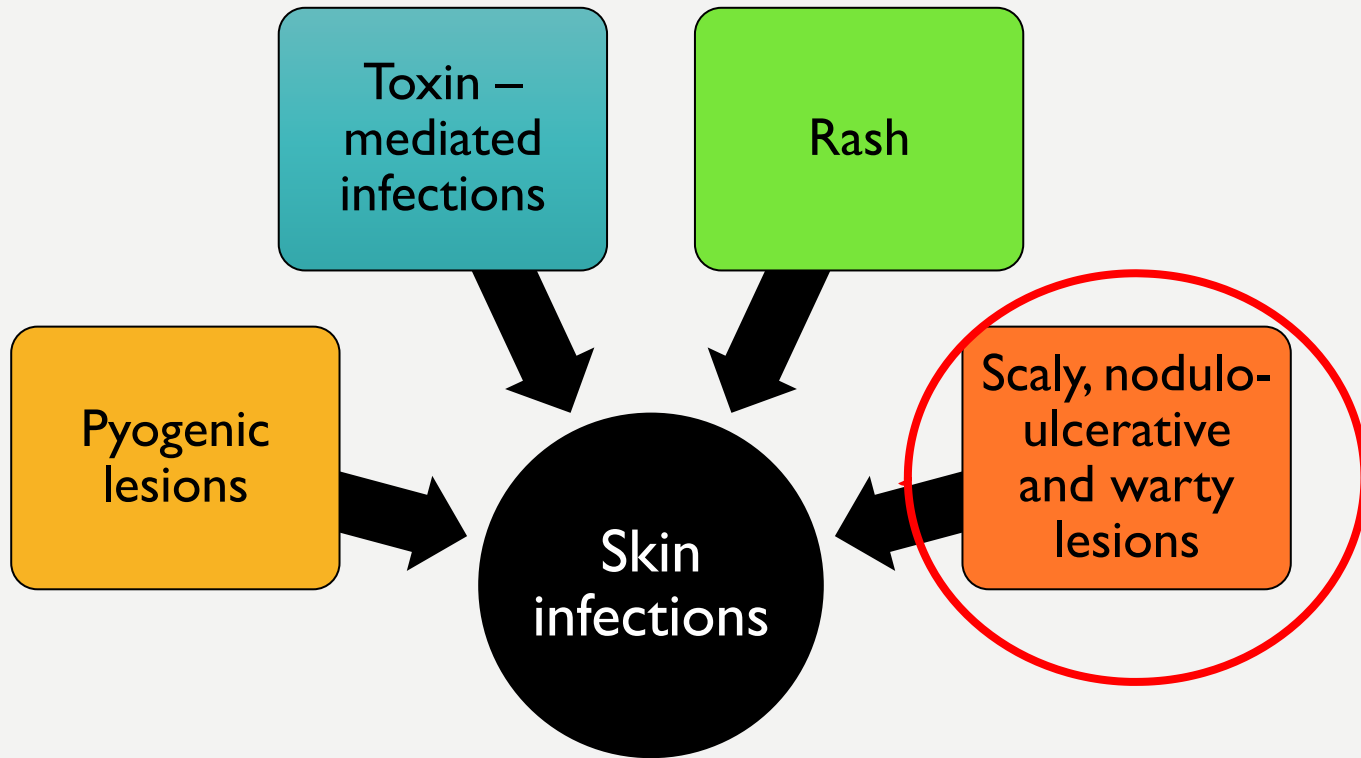
Bacterial infections:

1. Lyme Disease
2. Rocky Mountain Spotted Fever
3. Typhus fever
4. Leptospirosis
5. Meningococcal septicemia
6. Streptococcal- Scarlet fever

Recap- Lectures taught in
IS-FOM- Systemic infections
IS-MSN- Meningitis

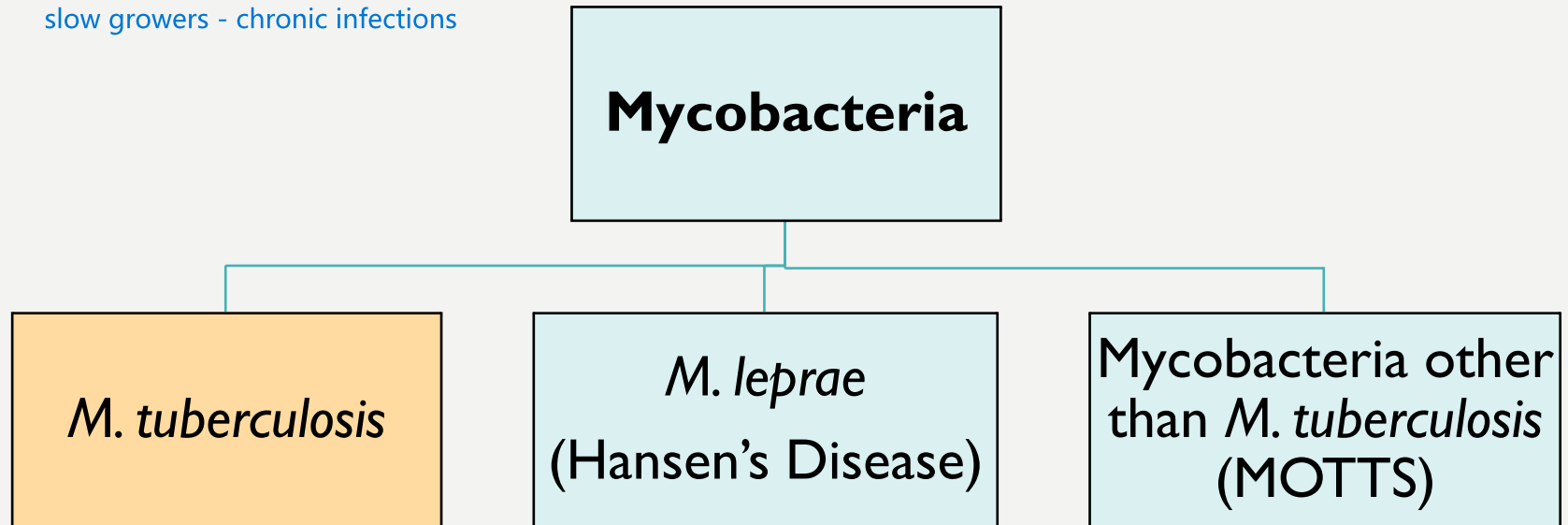
geographical locations, how to differentiate

TYPES OF SKIN INFECTIONS



***MYCOBACTERIUM* SPECIES**

slow growers - chronic infections



Acid fast stain
High lipid content (mycolic acid) in cell wall
Slow growing

gram stain isn't a good stain for these

MYCOBACTERIUM LEPRAE

- Acid-fast bacillus
- Obligate intracellular pathogen
 - No growth in artificial medium or tissue culture
 - Grown in animal models
- Reservoir
 - Human mucosa, skin, nerves
 - Grows better at lower than normal body temperature
- Causes leprosy (Hansen's Disease)
 - Rare in North America and Europe
 - Current endemic countries : Brazil, Madagascar, Tanzania, Mozambique and Nepal.
 - Chronic granulomatous disease of the peripheral nerves and superficial tissues, especially nasal mucosa



LEPROSY – PATHOGENESIS

Transmission- Person to person via inhalation of infectious aerosols or through skin contact with exudates from skin/ respiratory tract. requires prolonged contact

- **Pathogenesis**
 - **Obligate intracellular pathogen** of macrophages and Schwann cells
 - Does not produce toxins; **damage from host response**

- Two forms of leprosy

- Tuberculoid leprosy

Cell-mediated immune response → recovery

- Lepromatous leprosy

- **Antibody response, no/weak cell-mediated immune response → slowly progressive, fatal**



TYPES OF LEPROSY



Tuberculoid Leprosy

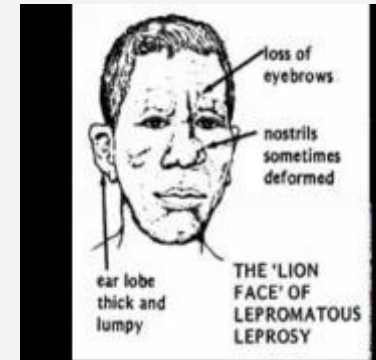
- Localized disease.
- Skin lesions are few erythematous/ hypo-pigmented plaques with flat centers.
- Peripheral nerve damage with complete sensory loss. Bigger nerve trunks thickened and deformities of hand and feet seen.
- **Bacilli in lesion- few or absent.**
- **Infectivity is low**
- **CMI- intact, TH1 response, IL-12**
- **Antibody levels low**
- **Lepromin test-POSITIVE**

sometimes hyperpigmented but more often hypo



Lepromatous Leprosy

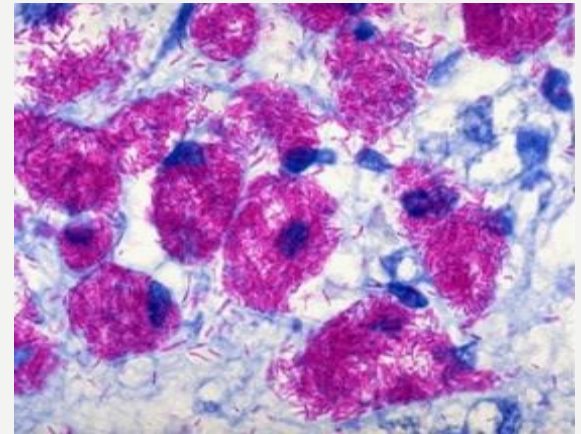
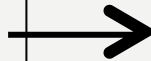
- Generalized disease.
- Numerous lesions -erythematous macules, papules or nodules, diffuse nerve involvement with patchy sensory loss, lack of nerve enlargement, destruction of nasal septum
- **Bacilli in lesion- numerous**
- **Infectivity is high.** immune system is weak - don't see cell mediated immune response
- **CMI- weak CMI, Th2 response, IL-4**
- **Antibody levels high**
- **Lepromin test-NEGATIVE**



LAB DIAGNOSIS AND TREATMENT

Lab diagnosis

Skin punch biopsy/ nasal scrapings-
Acid Fast Bacilli seen in clumps called globi within dermal macrophages and the Schwann cells.



Treatment

- Tuberculoid form
 - **Dapsone + rifampin**
- Lepromatous form
 - **Dapsone + rifampin + clofazimine**



Early treatment: before and after 6 months of therapy

NONTUBERCULOUS MYCOBACTERIA

AKA: MYCOBACTERIA OTHER THAN TUBERCULOSIS OR ATYPICAL MYCOBACTERIA

Noncontagious

- Acquired from environment
- Runyon Classification: differentiation based on production of **carotenoid pigments and growth rate** →
 - ❖ I. Photochromogens
 - Pigment only in light
 - ❖ II. Scotochromogens
 - Pigment in light and dark
 - ❖ III. Non-photochromogens
 - No pigment production
 - ❖ IV. Rapid growers
 - Colonies in <7 days



- *M. marinum* (Group I)
 - Soft tissue infection
 - “fish tank granuloma”

exposure to
water and fish



- *M. ulcerans* not common in US
- *M. avium-intracellulare* (Group III) and *M. kansasii* (Group I)
 - Respiratory, GI, disseminated
- *M. scrofulaceum* (group II)
 - From contaminated water
 - Cervical **lymphadenitis** in kids
can burst and ooze pus

NOCARDIOSIS-

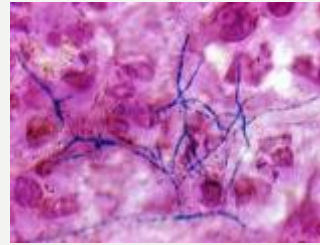
similar to actinomyces

Nocardia asteroides/ N. braziliensis

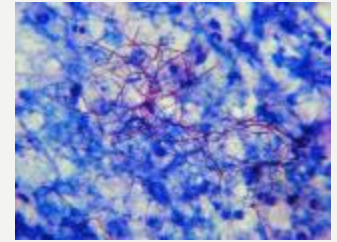
- The pathogenic nocardiae, like many nonpathogenic species of nocardia, are found worldwide in **soil and water**.
- Mode of infection: **Traumatic implantation**
- **Gram positive filamentous forms and partially Acid Fast**
- **Aerobic and** grow on a variety of media, producing heaped, **irregular, chalky colonies on Chocolate agar**
- They produce **urease and can digest paraffin**

Patient can present with:

- **Cellulitis/ swelling of tissues and which lead to subcutaneous abscesses with granules.**
- **Mycetoma - painless, chronic nodular lesions characterized by subcutaneous swelling, suppuration and formation of multiple sinus tracts.**



Gram positive
filamentous forms



Partially Acid Fast
bacilli in ZN stain



Lab diagnosis:

Tissue / abscess material

- **Microscopy**
Gram stain-branching ,
filamentous GPB
ZN stain-partially acid
fast filamentous bacilli.
- **Culture:** Chocolate
agar

Treatment:
Sulfonamides

Thank
you!!